Adhil Hamdan R I

Associate VLSI Engineer · DFT & Physical Design

+91-8903929373 · adhil30072003@gmail.com · linkedin.com/in/adhil-hamdan-aa98a724b · github.com/adhilhamdan · Bengaluru, India

SUMMARY

ECE graduate and Associate VLSI Engineer with industry exposure in Design for Testability at SanDisk (via Krivya Semicon), covering IP wrapper generation, IJTAG, MBIST insertion, and ATPG using Tessent and SpyGlass. Physical Design trained at MavenSilicon with end-to-end ASIC PD flow experience. Springer-indexed published researcher in image processing. Passionate about Silicon Lifecycle Management, DFT for low-power SoCs, and hardware-software co-design.

EXPERIENCE

Associate VLSI Engineer · Krivya Semicon (at SanDisk) — *Feb 2026 – Present*

- Involved in DFT workflows at SanDisk through Krivya as a service partner, contributing to IP wrapper generation, RTL creation, IJTAG concepts, padmuxing, MBIST insertion, and ATPG execution.
- Utilized industry-standard EDA tools including Siemens Tessent and Synopsys SpyGlass for DFT implementation and verification.

Physical Design Trainee · MavenSilicon — *Aug 2025 – Feb 2026*

- Completed end-to-end ASIC Physical Design training covering floorplanning, placement, Clock Tree Synthesis, routing, and Static Timing Analysis using Synopsys Fusion Compiler and PrimeTime.
- Executed independent project: full PD flow for a 1×3 Router, achieving timing closure with minimized skew and insertion delay.

Operations Associate Intern · Ninjacart — *Jan 2025 – Mar 2025*

- Optimized supply chain operations and inventory workflows by analysing large-scale datasets to improve fulfilment efficiency.
- Collaborated with Product, Analytics, and Operations teams using data-driven decision-making to identify and resolve bottlenecks.

Hardware Engineering Intern · Bright Burnishing Tools Pvt. Ltd. — *May 2024 – Jul 2024*

- Developed a real-time force monitoring solution using ESP32 with load sensors and signal conditioning circuits for industrial burnishing tools.
- Calibrated sensors against mechanical standards and analysed force-quality correlation to improve surface finish and tool performance.

Industrial Trainee · ISRO Propulsion Complex — *Dec 2023*

- Gained hands-on exposure in field instrumentation, flight instrumentation, control systems, measurement and data acquisition, and command systems for liquid propulsion systems.

EDUCATION

B.E. Electronics & Communication Engineering · PSG College of Technology — *Dec 2021 – Jul 2025*

CGPA: 7.93 · NIRF Rank: 67 · **Relevant Coursework:** Digital Electronics, Embedded Systems, VLSI Design, Analog Electronics, Operating Systems, Image Processing

PROJECTS

BIST for 4-bit ALU — *Jan 2024 – May 2024*

- Designed Built-In Self Test architecture using Galois LFSR for pattern generation and MISR for output compaction, achieving 100% fault coverage on a 4-bit ALU.
- Implemented complete RTL in Verilog and simulated using Xilinx Vivado 2018.1; published findings in IEEE-indexed conference paper.

1×3 Router Physical Design — *Sep 2025 – Dec 2025*

- Executed complete PD flow for a 1×3 router using Synopsys Fusion Compiler — floorplanning, placement, Clock Tree Synthesis, and detailed routing with optimized cell utilization.
- Performed timing closure using STA to resolve setup and hold violations, ensuring robust clock distribution with minimised skew and insertion delay.

Dual-Layer Image Security Using Chaotic Maps *Dec 2024 – Apr 2025*

- Designed end-to-end dual-layer image encryption framework in MATLAB using logistic and Henon chaotic maps for pixel scrambling and intensity transformation.
- Demonstrated enhanced resistance to statistical and differential attacks; presented at ICSSIC 2024 conference.

Adaptive DCP-Based Image Enhancement (MATLAB) *Jun 2024 – Nov 2024*

- Developed an Adaptive Dark Channel Prior framework integrated with fuzzy logic in MATLAB to reduce atmospheric haze for autonomous vehicle vision systems.
- Co-authored and published findings in a Springer-indexed journal (Lecture Notes in Networks and Systems).

Geofencing-Based Speed Limiting System *Jan 2023 – May 2023*

- Designed GPS-based speed control system using Haversine formula in microcontroller firmware, integrating GPS module, Hall Effect sensor, and servo motor for real-time speed restriction.
- Demonstrated automated audio/visual alert mechanism with high-accuracy zone detection.

Stroke Care 360 *Aug 2023 – May 2024*

- Engineered IoT smart pillbox module using microcontroller, RTC, and DFPlayer Mini for scheduled medication alerts as part of an assistive healthcare ecosystem.
- Implemented PPG signal processing to extract and monitor cardiovascular health metrics in real time; presented at Coimbatore Institute of Technology.

PUBLICATIONS

Adaptive Dark Channel Prior Based Image Enhancement for Weather Degraded Images

Springer — Lecture Notes in Networks and Systems · Oct 2025

DOI: https://link.springer.com/chapter/10.1007/978-3-031-99939-0_24

TECHNICAL SKILLS

Languages: Verilog, SystemVerilog, TCL, MATLAB, C, Python

VLSI Physical Design: Floorplanning, Placement, Clock Tree Synthesis (CTS), Routing, Static Timing Analysis (STA), Timing Closure, IR Drop Analysis, Physical Verification

DFT & Verification: Scan Insertion, ATPG, MBIST, IJTAG, IP Wrapper Generation, Fault Simulation, SpyGlass DFT, Tessent

EDA Tools: Synopsys Fusion Compiler, PrimeTime, Siemens Tessent, SpyGlass, ModelSim, Xilinx Vivado, Cadence EDA, OpenROAD

Embedded & Hardware: Arduino/AVR, ESP32, GPS/GSM Modules, Load Sensors, Signal Conditioning, PCB Prototyping, I2C/SPI/UART

Standards: IEEE 1149.1 (JTAG), IEEE 1687 (IJTAG), ISO 26262 basics, Silicon Lifecycle Management

LEADERSHIP & ACTIVITIES

- Founder and General Secretary, Green Kanyakumari Foundation — leading environmental and community initiatives. (Oct 2022 – Present)
- Secretary, IETE Student Chapter, PSG College of Technology — planned and executed technical events and student activities. (2022 – 2025)
- 1st Place, Nexus, IEEE ATOM 2024 — engineering competition. (Mar 2024)
- Danfoss Innovator Awards (Round 1) — developed a solution for minimising electricity theft. (Oct 2023)